

Why and How AI Systems Fail

WHY THIS MODULE MATTERS

Boards, employers and regulators are increasingly asking what leaders knew, and when. This module ensures you see AI risk early, understand your own exposure, and know what a failure actually looks like before it becomes one.

3.1 Three Categories of Failure

AI failures are often described narrowly as “bias” incidents, but the underlying causes are more varied. It is useful to separate failures into three broad categories: technical failures, where the system does not perform as designed; social failures, where the system performs as designed but produces harmful outcomes for people; and organisational failures, where the surrounding processes fail to catch a problem before it causes harm.

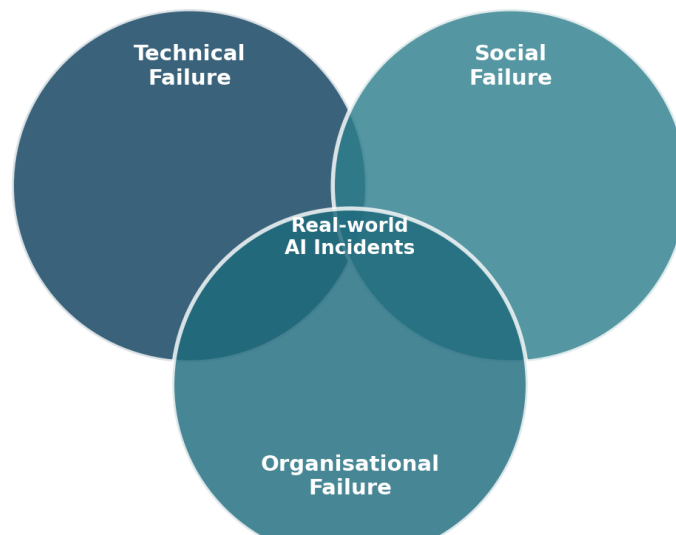


Figure 3.1 — Most documented AI incidents sit at the overlap of more than one failure category.

CASE STUDY — THE 2020 UK EXAM GRADING ALGORITHM

In March 2020, COVID-19 forced the cancellation of that summer's GCSE and A-level exams in England. Ofqual, the exams regulator, asked teachers to submit predicted grades and a ranking of their students, then applied a statistical standardisation algorithm intended to prevent grade inflation by aligning each school's results with its historical performance. When results were released on 13 August 2020, around 36% of A-level entries were downgraded by at least one grade from the teacher's prediction, and about 3% by two grades or more.

The algorithm relied heavily on a school's past results wherever a subject had more than 15 students, which systematically favoured small classes — disproportionately common at smaller, fee-paying schools — while larger comprehensive schools saw sharper downgrades. Analysis published after the fact found that students from disadvantaged backgrounds were roughly twice as likely to have their results downgraded as those from wealthier areas.

There was no meaningful, accessible appeals process in place when results were published; the one that existed required payment and was widely criticised as too complex under time pressure. Within two days of publication and mounting public protest, the Government reversed course and reverted to teachers' original predicted grades. Ofqual's chief regulator and the Department for Education's permanent secretary both resigned within weeks, and the Office for Statistics Regulation launched a review of the whole approach.

3.2 Technical Failure Modes

- **Data drift** — the statistical properties of real-world inputs shift away from the data a model was trained on, degrading accuracy silently over time.
- **Underspecified evaluation** — a model is validated against a narrow metric (such as overall accuracy) that hides poor performance on specific subgroups or edge cases. Ofqual's own technical report noted the algorithm had never been tested against real teacher-predicted rankings before it was used to grade real students.
- **Distribution mismatch** — training data does not represent the population or conditions the system encounters in deployment.
- **Brittleness** — a system performs well on typical inputs but degrades sharply on unusual, adversarial or out-of-scope inputs it was never tested against — small subject classes were exactly the edge case Ofqual's algorithm handled worst.

3.3 Social Failure Modes

Social failures occur when a system works exactly as intended, but the intention itself, or its application, causes harm. Ofqual's algorithm largely achieved its stated technical goal — a grade distribution similar to previous years — while systematically disadvantaging state-educated students in larger year groups, which is precisely why achieving the intended technical target is not the same thing as achieving a fair outcome.

- **Proxy discrimination** — a model uses a variable correlated with a protected or disadvantaged characteristic as an effective stand-in for it; school-level historical performance acted as a strong proxy for socioeconomic background.

- **Feedback loops** — a system's own outputs shape future data, reinforcing an initial bias over successive cycles.
- **Automation bias** — human reviewers defer to a system's output even when their own judgment suggests otherwise, eroding the value of human oversight.
- **Misaligned optimisation** — the metric a system is trained to maximise (grade-distribution consistency with prior years) diverges from the outcome the public and students actually needed (an individually fair, defensible grade).

3.4 Organisational Failure Modes

Even a well-designed system can cause harm if the organisation around it lacks the structures to notice and respond to problems. Ofqual tested eleven candidate algorithms against 2019 results before choosing one, which shows technical diligence — but no equivalent diligence went into stress-testing the public communication, the appeals route, or what would happen politically if the model's own trade-offs, which were known internally in advance, became visible to the public all at once.

Notable Patterns in UK Public Sector AI

Independent reviews of automated and algorithmic tools used across UK central and local government have repeatedly identified a similar pattern to Ofqual's: tools deployed without a clear public-facing owner, limited published documentation of how a tool reaches its outputs, and no accessible route for someone affected by a decision to challenge it before the decision takes effect. These are organisational failures as much as technical ones, and they are precisely what the governance structures in Module 5 are designed to prevent.

TRY THIS — CLASSIFY THE FAILURE

Using the three-category framework above, identify at least one technical, one social, and one organisational failure in the Ofqual case specifically — not in general terms, but citing the specific fact from the case study that supports each classification. (A model answer: technical — the algorithm was untested against real ranking data before use; social — school history acted as a proxy for disadvantage; organisational — no working appeals route existed at launch, despite the trade-offs being known internally in advance.)

3.5 The Five Categories of AI Risk

Boards and senior leaders are rarely well served by thinking about “AI risk” as a single, undifferentiated concern. It is more useful to track five distinct categories, each of which calls for a different owner and a different set of controls. Ofqual's episode touched all five simultaneously: operational (results had to be reissued within days), legal (subsequent legal challenges were prepared against the regulator), strategic (an entire year's admissions cycle was thrown into doubt), reputational (two senior officials resigned), and, in later systems built on similar logic, cyber risk around the integrity of the underlying data feeding the model.



Figure 3.2 — Five categories of AI risk that every board should track separately.

- **Operational risk** — the system fails, degrades or behaves unpredictably in ways that disrupt normal business activity.
- **Legal risk** — the system's use breaches a statutory duty, a regulator's expectations, or a contractual obligation to a client or supplier.
- **Strategic risk** — an over- or under-investment in AI relative to competitors, or a governance failure that blocks a genuinely valuable use case.
- **Reputational risk** — the system's behaviour, or the organisation's response to a failure, damages trust with customers, staff, regulators or the public.
- **Cyber risk** — the system introduces a new attack surface, or is itself used as a tool for attack — including deepfakes, data poisoning and prompt injection.

IN PRACTICE

A logistics firm's delivery-routing AI starts producing noticeably worse routes after a software update from its vendor. Logged against the five categories: operational (drivers are taking longer routes), legal (delivery-time contracts with clients may be breached), reputational (customers are complaining publicly). Naming all three separately, rather than just “the AI is broken”, gets the right people — operations, legal, and customer service — into the same conversation on day one instead of days three or four.

3.6 Failure Patterns Specific to Data Centre Operations

The AI systems that manage a modern data centre's physical operation have their own well-documented failure patterns, and they map onto the same three categories introduced above.

- **Technical** — predictive cooling and power-load models trained on historical demand can under-forecast a sudden spike in AI training workloads, contributing to thermal events; predictive maintenance models can miss failure modes not present in their training history, producing false negatives on ageing hardware.
- **Social** — biometric access control systems with uneven accuracy across demographic groups can create a recurring, demoralising experience for some staff, who are disproportionately stopped, delayed or flagged at secure entry points.
- **Organisational** — responsibility for an AI-driven building management system is often split between a facilities team, an IT/OT team and an external vendor, with no single named owner — precisely the diffusion of accountability described in section 3.4.

Because these systems sit at the intersection of physical safety, security and IT, a fault can escalate unusually quickly — a cooling failure is not just a technical bug, it is a site incident with potential customer, safety and reputational consequences within minutes, not weeks. This is why Module 9 treats incident response as a distinct, higher-tempo governance discipline within the data centre environment.

KEY TAKEAWAYS

- AI failures fall into technical, social and organisational categories — and most real incidents, including the 2020 Ofqual grading algorithm, involve more than one at once.
- A system can hit its own stated technical target — a stable grade distribution — while still producing a socially and politically unacceptable outcome.
- Organisational gaps — unclear ownership, weak monitoring, no working appeals route — turn a technical or social issue into a prolonged, undetected harm.
- Operational, legal, strategic, reputational and cyber risk are distinct categories that deserve separate owners and controls, not a single undifferentiated “AI risk” line.
- Data centre building-management and access-control systems fail in the same three ways, but the physical and safety stakes mean faults can escalate into site incidents very quickly.

FURTHER READING

- Ofqual — technical report on the 2020 exam grading standardisation model
- Office for Statistics Regulation — review of the 2020 exams episode
- ProPublica — “Machine Bias” and “How We Analyzed the COMPAS Recidivism Algorithm”, [propublica.org](https://www.propublica.org)

MODULE 3 CHECK-IN QUIZ

Five questions to confirm your understanding before moving on. Answers are provided in the Answer Key at the end of the course.

1. Which of the following is an example of a technical AI failure mode?

- A. Automation bias among human reviewers
- B. Data drift degrading model accuracy over time
- C. Unclear ownership of a deployed system
- D. No escalation path for affected users

2. A model that is statistically accurate on average but systematically harms an underrepresented subgroup is best described as which type of failure?

- A. Purely technical
- B. Purely organisational
- C. Social
- D. Not a failure at all

3. What is a “feedback loop” in the context of AI failure?

- A. A user survey conducted after deployment
- B. A system's own outputs shaping future data in ways that reinforce an initial bias
- C. A formal escalation channel for complaints
- D. A scheduled model retraining cycle

4. According to Module 3, what pattern have independent reviews of UK public sector AI tools repeatedly identified?

- A. Excessive human oversight slowing down decisions
- B. Unclear post-deployment ownership and limited routes for affected people to challenge decisions
- C. Overly cautious data minimisation
- D. Too many named accountable owners

5. Why do most serious AI incidents tend to involve more than one failure category at once?

- A. Because technical, social and organisational weaknesses often compound together
- B. Because regulators require multiple categories to be reported
- C. Because AI systems always fail for identical reasons
- D. Because organisational failures are impossible to prevent